Math 54 — Week 5

Properties of determinants; Cramer's rule and volume; vector spaces; null and column spaces (Lay 3.2–3.3, 4.1–4.2)

§1 Facts used from Weeks 1–4

(W1) $\det A$ by cofactor expansion along any row or column; for a triangular matrix, the product of the diagonal entries.

(W2) An elementary matrix E performs a row operation: EA is A after that operation.

(W3) Invertible matrix theorem: for square A, invertible $\iff$ n pivots $\iff$ $A\mathbf{x} = \mathbf{0}$ only trivially $\iff$ columns span $\mathbb{R}^n$, etc.

(W4) The solution set of $A\mathbf{x} = \mathbf{0}$ is the span of the vectors in its parametric vector form. $A\mathbf{x} = \mathbf{b}$ is consistent $\iff$ $\mathbf{b}$ is a combination of the columns of A.

§2 This week

§2.1 Properties of determinants (Lay 3.2)

The question. Cofactor expansion of a 10×10 matrix takes millions of multiplications. Row reduction takes a few hundred. How does each row operation change the determinant, so that row reduction can compute it?

Theorem (Row operations and the determinant)

Let A be square.

(a) If a multiple of one row is added to another row to give B, then $\det B = \det A$.

(b) If two rows are interchanged to give B, then $\det B = -\det A$.

(c) If one row is multiplied by k to give B, then $\det B = k \det A$.

The idea

Start with (c): expand along the scaled row; every term of the expansion picks up the factor k. For (b) and (a), check 2×2 first: swapping the rows of the matrix with rows (a, b) and (c, d) gives $cb - da = -(ad - bc)$; adding k times row 1 to row 2 gives $a(d + kb) - b(c + ka) = ad - bc$. For larger matrices, expand along a row that the operation did not touch: each cofactor is the determinant of a smaller matrix on which the same operation was performed, so the claim for size n follows from the claim for size $n - 1$ (induction). (For $n \geq 3$ every one of these operations involves at most two rows, so an untouched row exists.)

Consequences.

- **Determinant by row reduction**: reduce A to an echelon form U using only replacements and r interchanges. U is triangular, so
$$\det A = (-1)^r \cdot (\text{product of the diagonal entries of } U).$$

- Example:
$$\begin{bmatrix} 1 & -4 & 2 \\ -2 & 8 & -9 \\ -1 & 7 & 0 \end{bmatrix} \xrightarrow[R_3+R_1]{R_2+2R_1} \begin{bmatrix} 1 & -4 & 2 \\ 0 & 0 & -5 \\ 0 & 3 & 2 \end{bmatrix} \xrightarrow{R_2 \leftrightarrow R_3} \begin{bmatrix} 1 & -4 & 2 \\ 0 & 3 & 2 \\ 0 & 0 & -5 \end{bmatrix}.$$
One interchange, so $\det A = -(1)(3)(-5) = 15$.
- A matrix with two equal rows, or a zero row, has determinant 0 (subtract one equal row from the other to get a zero row; expand along a zero row).

Theorem (Invertible $\iff$ nonzero determinant)
A square matrix A is invertible if and only if $\det A \neq 0$.

Proof. In the formula $\det A = (-1)^r \cdot$(product of the diagonal of U), the diagonal of a square echelon form U consists of the pivots if there are n of them; otherwise the last diagonal entry is 0. So $\det A \neq 0 \iff U$ has n pivots $\iff A$ is invertible (W3). □

This adds a statement to the invertible matrix theorem: $\det A \neq 0$.

Theorem (Transpose and product)
$\det A^T = \det A$, and $\det(AB) = (\det A)(\det B)$ for $n \times n$ matrices A, B.

The idea

$\det A^T = \det A$: expanding A^T along a row is expanding A along a column, and the cofactor expansion theorem allows both (induction on n). It follows that *column* operations change the determinant exactly as row operations do.

$\det(AB)$: if A is not invertible, neither is AB (if AB had an inverse C, then $A(BC) = I$, making A invertible), so both sides are 0. If A is invertible, write $A = E_p \cdots E_1$ as a product of elementary matrices. By the row-operation theorem, $\det(EM) = \det(E)\det(M)$ for every elementary E (with $\det E = 1$, -1 or k); apply this p times to both AB and A.

- Consequences: $\det(A^{-1}) = 1/\det A$ (from $\det(A)\det(A^{-1}) = \det I = 1$); $\det(A^k) = (\det A)^k$; $\det(cA) = c^n \det A$ for $n \times n$ A (each of the n rows is scaled).
- $\det(A+B) \neq \det A + \det B$ in general.
- **Linearity in one column**: with all other columns fixed, $\mathbf{x} \mapsto \det[\,\mathbf{a}_1 \cdots \mathbf{x} \cdots \mathbf{a}_n\,]$ is a linear function of $\mathbf{x}$ (expand down that column).

§2.2 Cramer's rule, the inverse formula, and volume (Lay 3.3)

Notation. $A_i(\mathbf{b})$ is A with column i replaced by $\mathbf{b}$.

Theorem (Cramer's rule)
If A is invertible $n \times n$, the unique solution of $A\mathbf{x} = \mathbf{b}$ has entries $x_i = \frac{\det A_i(\mathbf{b})}{\det A}$.

The idea

The identity matrix with column i replaced by $\mathbf{x}$, namely $I_i(\mathbf{x})$, has determinant x_i: expand along row i, whose only nonzero entry is x_i in the diagonal position, and deleting row i and column i leaves I_{n-1}. Multiplying by A turns each column $\mathbf{e}_j$ into $\mathbf{a}_j$ and the column $\mathbf{x}$ into $A\mathbf{x} = \mathbf{b}$. So $AI_i(\mathbf{x}) = A_i(\mathbf{b})$. Take determinants.

Proof.
$$AI_i(\mathbf{x}) = [\,A\mathbf{e}_1 \cdots A\mathbf{x} \cdots A\mathbf{e}_n\,] = [\,\mathbf{a}_1 \cdots \mathbf{b} \cdots \mathbf{a}_n\,] = A_i(\mathbf{b}).$$
So $(\det A)(\det I_i(\mathbf{x})) = \det A_i(\mathbf{b})$, and $\det I_i(\mathbf{x}) = x_i$ (expand along row i). Divide by $\det A \neq 0$. □

Example: $2x_1 + x_2 = 7$, $-3x_1 + 4x_2 = 6$.

$$\det A = \det \begin{bmatrix} 2 & 1 \\ -3 & 4 \end{bmatrix} = 11, \quad \det A_1(\mathbf{b}) = \det \begin{bmatrix} 7 & 1 \\ 6 & 4 \end{bmatrix} = 22, \quad \det A_2(\mathbf{b}) = \det \begin{bmatrix} 2 & 7 \\ -3 & 6 \end{bmatrix} = 33,$$

so $x_1 = 22/11 = 2$ and $x_2 = 33/11 = 3$.

Theorem (Inverse formula)
If A is invertible, $A^{-1} = \frac{1}{\det A} \operatorname{adj} A$, where the **adjugate** $\operatorname{adj} A$ is the *transpose* of the matrix of cofactors: $(\operatorname{adj} A)_{ij} = C_{ji}$.

Proof. Column j of A^{-1} solves $A\mathbf{x} = \mathbf{e}_j$. By Cramer's rule its entry i is $\det A_i(\mathbf{e}_j)/\det A$, and expanding $A_i(\mathbf{e}_j)$ down column i (a single 1 in row j) gives $(-1)^{i+j} \det A_{ji} = C_{ji}$. □

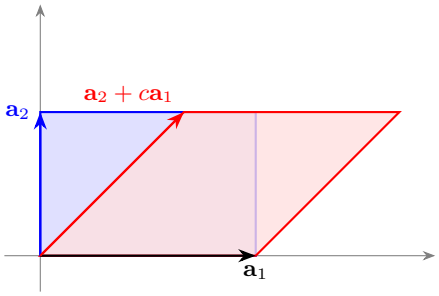
Useful mainly for 2×2 (it gives the formula from Week 4) and for theory; for numbers, row reduce $[\,A \;\; I\,]$.

Determinants as area and volume.

Theorem (Area and volume)
If A is 2×2, $|\det A|$ is the area of the parallelogram determined by the columns of A. If A is 3×3, $|\det A|$ is the volume of the parallelepiped determined by the columns.

The idea

For a diagonal matrix the parallelogram is a rectangle with sides $|a|$ and $|d|$, and $|\det| = |ad|$. A column replacement $\mathbf{a}_2 \to \mathbf{a}_2 + c\mathbf{a}_1$ slides the parallelogram along the direction of $\mathbf{a}_1$: same base, same height, same area; and it does not change det either (column version of (a)). Swapping columns changes neither the area nor $|\det|$. Column operations reduce any matrix to diagonal form (or produce a zero column, area 0, $\det 0$), so area and $|\det|$ agree.



Same base $\mathbf{a}_1$, same height: the slanted parallelogram has the same area as the rectangle, and the column replacement did not change det.

Why area equals $|\det A|$.

- A parallelogram with vertices $\mathbf{p}, \mathbf{p}+\mathbf{u}, \mathbf{p}+\mathbf{v}, \mathbf{p}+\mathbf{u}+\mathbf{v}$: translate $\mathbf{p}$ to the origin; area $= |\det[\,\mathbf{u} \;\; \mathbf{v}\,]|$.
- **Linear transformations scale area by** $|\det A|$: if $T(\mathbf{x}) = A\mathbf{x}$ and S is a parallelogram (or any region with finite area) in $\mathbb{R}^2$, then $\text{area}(T(S)) = |\det A| \cdot \text{area}(S)$. In $\mathbb{R}^3$, the same with volume. Example: the ellipse $x_1^2/a^2 + x_2^2/b^2 \le 1$ is the image of the unit disk under

$$\begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix},$$

so its area is πab.

§2.3 Vector spaces and subspaces (Lay 4.1)

The question. Polynomials, functions and matrices can all be added and scaled like vectors in $\mathbb{R}^n$. What exactly is needed for the ideas of span and linear combination to work for them too?

Definition. A **vector space** is a nonempty set V with an addition and a multiplication by real scalars such that, for all $\mathbf{u}, \mathbf{v}, \mathbf{w}$ in V and scalars c, d:

(1) $\mathbf{u}+\mathbf{v}$ is in V;
(2) $\mathbf{u}+\mathbf{v}=\mathbf{v}+\mathbf{u}$;
(3) $(\mathbf{u}+\mathbf{v})+\mathbf{w}=\mathbf{u}+(\mathbf{v}+\mathbf{w})$;
(4) there is a $\mathbf{0}$ in V with $\mathbf{u}+\mathbf{0}=\mathbf{u}$;
(5) each $\mathbf{u}$ has a $-\mathbf{u}$ with $\mathbf{u}+(-\mathbf{u})=\mathbf{0}$;
(6) $c\mathbf{u}$ is in V;
(7) $c(\mathbf{u}+\mathbf{v})=c\mathbf{u}+c\mathbf{v}$;
(8) $(c+d)\mathbf{u}=c\mathbf{u}+d\mathbf{u}$;
(9) $c(d\mathbf{u})=(cd)\mathbf{u}$;
(10) $1\mathbf{u}=\mathbf{u}$.

- Examples: $\mathbb{R}^n$; $\mathbb{P}_n$, the polynomials of degree at most n; all real-valued functions on a set; the $m \times n$ matrices. In each, the operations are the familiar ones, and the axioms reduce to rules for numbers.
- From the axioms: $0\mathbf{u}=\mathbf{0}$, $c\mathbf{0}=\mathbf{0}$, $-\mathbf{u}=(-1)\mathbf{u}$.

Definition. A **subspace** of V is a subset H with: (a) $\mathbf{0} \in H$; (b) $\mathbf{u}+\mathbf{v} \in H$ whenever $\mathbf{u}, \mathbf{v} \in H$; (c) $c\mathbf{u} \in H$ whenever $\mathbf{u} \in H$. A subspace is itself a vector space (the other axioms are inherited from V).

- $\{\mathbf{0}\}$ and V are subspaces. A line through the origin in $\mathbb{R}^2$ is a subspace; a line not through the origin is not (it misses $\mathbf{0}$). $\mathbb{R}^2$ is *not* a subspace of $\mathbb{R}^3$ (its vectors have two entries, not three); the plane $\{(s,t,0)\}$ in $\mathbb{R}^3$ is.
- $\{p \in \mathbb{P}_2 : p(1)=0\}$ is a subspace of $\mathbb{P}_2$: the zero polynomial vanishes at 1, and sums and multiples of polynomials vanishing at 1 vanish at 1.

Theorem (A span is a subspace)
If $\mathbf{v}_1, \ldots, \mathbf{v}_p$ are in V, then $\operatorname{Span}\{\mathbf{v}_1, \ldots, \mathbf{v}_p\}$ is a subspace of V.

Proof. $\mathbf{0}=0\mathbf{v}_1+\cdots+0\mathbf{v}_p$. The sum of $\sum c_i\mathbf{v}_i$ and $\sum d_i\mathbf{v}_i$ is $\sum(c_i+d_i)\mathbf{v}_i$, and $k\sum c_i\mathbf{v}_i=\sum(kc_i)\mathbf{v}_i$; both are again combinations. □

Shortcut for subspace questions: if you can write H as a span (or, below, as a null space or column space), it is a subspace with no axioms to check.

§2.4 Null spaces, column spaces, row spaces (Lay 4.2)

Definition. For an $m \times n$ matrix A: the **null space** $\operatorname{Nul} A = \{\mathbf{x} \in \mathbb{R}^n : A\mathbf{x}=\mathbf{0}\}$; the **column space** $\operatorname{Col} A = \operatorname{Span}\{\text{columns of } A\} \subseteq \mathbb{R}^m$; the **row space** $\operatorname{Row} A = \operatorname{Span}\{\text{rows of } A\}$ (rows written as vectors in $\mathbb{R}^n$).

Theorem (Null spaces and column spaces are subspaces)
$\operatorname{Nul} A$ is a subspace of $\mathbb{R}^n$ and $\operatorname{Col} A$ is a subspace of $\mathbb{R}^m$.

Proof. $A\mathbf{0}=\mathbf{0}$. If $A\mathbf{u}=\mathbf{0}$ and $A\mathbf{v}=\mathbf{0}$, then $A(\mathbf{u}+\mathbf{v})=\mathbf{0}$ and $A(c\mathbf{u})=\mathbf{0}$. $\operatorname{Col} A$ is a span. □

An explicit description of $\operatorname{Nul} A$**.** Solve $A\mathbf{x}=\mathbf{0}$ in parametric vector form; the vectors attached to the free variables span $\operatorname{Nul} A$. For example:

$$A=\begin{bmatrix}1&-2&0&-1&3\\0&0&1&2&-2\\2&-4&1&0&4\end{bmatrix} \longrightarrow \operatorname{RREF} \begin{bmatrix}1&-2&0&-1&3\\0&0&1&2&-2\\0&0&0&0&0\end{bmatrix}.$$

So $x_1=2x_2+x_4-3x_5$ and $x_3=-2x_4+2x_5$, with x_2, x_4, x_5 free, and

$$\operatorname{Nul} A=\operatorname{Span}\left\{\begin{bmatrix}2\\1\\0\\0\\0\end{bmatrix},\begin{bmatrix}1\\0\\-2\\1\\0\end{bmatrix},\begin{bmatrix}-3\\0\\2\\0\\1\end{bmatrix}\right\}.$$

Each of these vectors has a 1 in its own free-variable slot and 0 in the other free slots, so none is a combination of the others: the spanning set produced this way is always linearly independent.

$\operatorname{Nul} A$	$\operatorname{Col} A$
subspace of $\mathbb{R}^n$ (inputs)	subspace of $\mathbb{R}^m$ (outputs)
defined implicitly: a condition $A\mathbf{x} = \mathbf{0}$	defined explicitly: you can write down members
finding members takes row reduction	the columns are members; no work
$\mathbf{v} \in \operatorname{Nul} A$? Compute $A\mathbf{v}$.	$\mathbf{v} \in \operatorname{Col} A$? Row reduce $[\,A \;\; \mathbf{v}\,]$.
$\operatorname{Nul} A = \{\mathbf{0}\} \iff A\mathbf{x} = \mathbf{0}$ only trivially	$\operatorname{Col} A = \mathbb{R}^m \iff$ pivot in every row

Linear transformations between vector spaces. $T\colon V \to W$ is linear if $T(\mathbf{u}+\mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$ and $T(c\mathbf{u}) = cT(\mathbf{u})$. Its **kernel** $\{\mathbf{u} : T(\mathbf{u}) = \mathbf{0}\}$ and **range** $\{T(\mathbf{x})\}$ are subspaces of V and W. For $T(\mathbf{x}) = A\mathbf{x}$ they are $\operatorname{Nul} A$ and $\operatorname{Col} A$. Example: differentiation $D\colon \mathbb{P}_3 \to \mathbb{P}_2$, $D(p) = p'$, is linear; its kernel is the constant polynomials.

Shortcuts

- **Determinant by row reduction**: only replacements and swaps; count the swaps; multiply the diagonal. Factor out a common factor of a row first if it saves fractions (it multiplies det).
- **Combine expansion and reduction**: create zeros in one column with replacements, then expand down that column.
- **Determinant rules**: $\det(AB) = \det A \det B$, $\det A^{-1} = 1/\det A$, $\det A^T = \det A$, $\det(cA) = c^n \det A$.
- **Invertible for which h?** Compute det as a polynomial in h and find its zeros.
- **Cramer's rule** is fastest for 2×2, or for one single entry x_i of a 3×3 solution.
- **Area of a triangle** with vertices $\mathbf{0}, \mathbf{u}, \mathbf{v}$: $\frac{1}{2}|\det[\,\mathbf{u}\;\mathbf{v}\,]|$.
- **Is H a subspace?** First check $\mathbf{0} \in H$; if not, stop. If H is described by homogeneous linear equations, it is a null space; if it is "all vectors of the form ..." with linear expressions in parameters, it is a span.
- **Spanning set for** $\operatorname{Nul} A$: one vector per free variable, from the parametric vector form.

Traps

- Scaling a row by k multiplies det by k; when you scale during row reduction, divide it back out.
- $\det(cA) = c^n \det A$, not $c \det A$. $\det(A+B) \neq \det A + \det B$.
- The adjugate is the *transpose* of the cofactor matrix.
- Area is $|\det|$; the sign of det records orientation.
- A set defined by $x_1 + x_2 = 1$, or by $x_1 x_2 \geq 0$, is not a subspace (first misses $\mathbf{0}$; second is not closed under addition: $(1,0) + (0,-1)$).
- $\operatorname{Nul} A$ lives in $\mathbb{R}^n$, $\operatorname{Col} A$ in $\mathbb{R}^m$: different spaces when $m \neq n$.

§3 Homework this week

HW4, Section 3.2: 2, 3, 5, 9, 15, 19, 20, 23, 24, 27, 29, 31. Section 3.3: 3, 5, 9, 11, 21, 27, 29, 30. Section 4.1: 2, 3, 6, 7, 10, 11, 20, 23, 28, 32. HW5, Section 4.2: 1, 5, 9, 13, 15, 17, 24, 25, 31, 32, 36, 38, 40, 43, 46.

§4 Problem set

P1. Compute

$$\det \begin{bmatrix} 2 & 4 & -2 & 6 \\ 1 & 3 & 0 & 2 \\ -1 & -1 & 4 & 1 \\ 3 & 8 & 1 & 5 \end{bmatrix}$$

by row reduction.

P2. A and B are 3×3 with $\det A = 3$ and $\det B = -2$. Find $\det(AB)$, $\det(A^{-1})$, $\det(A^T)$, $\det(2A)$, $\det(B^T A^{-1} B)$.

P3. For which h is

$$\begin{bmatrix} 1 & 2 & h \\ 2 & 5 & 1 \\ 0 & 1 & 3 \end{bmatrix}$$

invertible?

P4. Use Cramer's rule to solve $5x_1 + 2x_2 = 1$, $3x_1 + x_2 = 4$.

P5. Find the area of the parallelogram with vertices $(-1, 0)$, $(0, 5)$, $(1, -4)$, $(2, 1)$. Then find the area of its image under

$$T(\mathbf{x}) = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix} \mathbf{x}.$$

P6. Which are subspaces? (a) $\{(a, b, c) \in \mathbb{R}^3 : a + b = 0\}$. (b) $\{(a, b) \in \mathbb{R}^2 : ab \geq 0\}$. (c) $\{p \in \mathbb{P}_2 : p(0) = 1\}$. (d) $\{(s - 2t,\ s,\ 3t)\}$, s, t real. (e) The set of 2×2 matrices A with $A^T = A$.

P7. For

$$A = \begin{bmatrix} 1 & -3 & -4 \\ -4 & 6 & -2 \\ -3 & 7 & 6 \end{bmatrix}:$$

find a spanning set for $\operatorname{Nul} A$, and decide whether $\mathbf{b} = (3, 3, -4)$ is in $\operatorname{Col} A$.

P8. For

$$A = \begin{bmatrix} 2 & 4 & -2 & 1 \\ -2 & -5 & 7 & 3 \\ 3 & 7 & -8 & 6 \end{bmatrix},$$

$\mathbf{u} = (3, -2, -1, 0)$, $\mathbf{v} = (3, -1, 3)$: is $\mathbf{u}$ in $\operatorname{Nul} A$? Is $\mathbf{v}$ in $\operatorname{Col} A$? Could $\mathbf{u}$ be in $\operatorname{Col} A$?